

QUESTION 1 | Integral Calculus — Linear Substitution

[2 Marks]

Question: Find $\int \sqrt{4x+1} \, dx$.**NESA Reference Sheet / Key Rule:**

$$\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C, \quad \text{where } n \neq -1$$

Detailed Working (NESA Standard Solution):

1. Express the integrand in fractional index form:

$$\int \sqrt{4x+1} \, dx = \int (4x+1)^{\frac{1}{2}} \, dx$$

2. Identify linear parameters: $a = 4$, $b = 1$, and power $n = \frac{1}{2}$.

3. Apply the linear composite primitive rule:

$$\int (4x+1)^{\frac{1}{2}} \, dx = \frac{(4x+1)^{\frac{1}{2}+1}}{4 \cdot (\frac{1}{2}+1)} + C = \frac{(4x+1)^{\frac{3}{2}}}{4 \cdot \frac{3}{2}} + C = \frac{1}{6}(4x+1)^{\frac{3}{2}} + C$$

Final Answer: $\frac{1}{6}(4x+1)^{\frac{3}{2}} + C$ or $\frac{\sqrt{(4x+1)^3}}{6} + C$

Marks	Marking Criteria Breakdown
1 Mark	Expresses integrand in index form and applies linear rule (divides by 4 and adds 1 to power).
1 Mark	Simplifies denominator to 6 and includes constant of integration + C.

QUESTION 2 | Differential Calculus — Increasing Functions

[2 Marks]

Question: For the function $f(x) = x^3 - 9x$, state the domain where $f(x)$ is increasing.**NESA Reference Sheet / Key Rule:**A differentiable function $f(x)$ is increasing on an interval where $f'(x) \geq 0$ (or strictly increasing where $f'(x) > 0$).**Detailed Working (NESA Standard Solution):**

1. Find the first derivative:

$$f'(x) = \frac{d}{dx}(x^3 - 9x) = 3x^2 - 9$$

2. Set up the condition for an increasing function:

$$f'(x) \geq 0 \Rightarrow 3x^2 - 9 \geq 0 \Rightarrow 3(x^2 - 3) \geq 0 \Rightarrow x^2 \geq 3$$

3. Solve the quadratic inequality:

$$x \leq -\sqrt{3} \quad \text{or} \quad x \geq \sqrt{3} \quad \text{or in interval notation: } (-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$$

*(Note: NESA marking criteria accepts both non-strict $x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ and strict $x < -\sqrt{3}$ or $x > \sqrt{3}$).***Final Answer:** $x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ $((-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty))$

Marks	Marking Criteria Breakdown
1 Mark	Differentiates correctly to obtain $f'(x) = 3x^2 - 9$.
1 Mark	Solves inequality $f'(x) \geq 0$ to state exact domain in terms of $\pm\sqrt{3}$.

QUESTION 3

Integral Calculus — Logarithmic Integrals

[2 Marks]

Question: Find $\int \frac{e^{3x}}{e^{3x}+1} \, dx$.

NESA Reference Sheet / Key Rule:
 $\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + c$

Detailed Working (NESA Standard Solution):

1. Recognise that the numerator is a scalar multiple of the derivative of the denominator:

$$\text{Let } f(x) = e^{3x} + 1 \Rightarrow f'(x) = 3e^{3x}$$

2. Balance the scalar multiplier:

$$\int \frac{e^{3x}}{e^{3x} + 1} \, dx = \frac{1}{3} \int \frac{3e^{3x}}{e^{3x} + 1} \, dx$$

3. Integrate using the standard logarithmic primitive:

$$= \frac{1}{3} \ln(e^{3x} + 1) + C$$

(Absolute value signs are not required since $e^{3x} + 1 > 0$ for all real x).

Final Answer: $\frac{1}{3} \ln(e^{3x} + 1) + C$

Marks	Marking Criteria Breakdown
1 Mark	Recognises the $\frac{f'(x)}{f(x)}$ form and balances with constant factor $\frac{1}{3}$.
1 Mark	Obtains correct primitive including constant of integration $+ C$.

QUESTION 4 | Calculus — Chain Rule & Reverse Chain Rule

[4 Marks]

Question: a. Find $\frac{d}{dx}(\sqrt{x} + 1)^3$
b. Hence, find $\int \frac{(\sqrt{x}+1)^2}{2\sqrt{x}} dx$

NESA Reference Sheet / Key Rule:
Chain Rule: $\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$

Detailed Working (NESA Standard Solution):

Part a (Differentiating via Chain Rule):

- Let $g(x) = \sqrt{x} + 1 = x^{\frac{1}{2}} + 1 \Rightarrow g'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$.
- Differentiating:

$$\frac{d}{dx}[(\sqrt{x} + 1)^3] = 3(\sqrt{x} + 1)^2 \cdot \frac{1}{2\sqrt{x}} = \frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}}$$

Part b (Using ‘Hence’ to integrate):

- From Part (a), we have the exact differential relationship:

$$\frac{d}{dx}[(\sqrt{x} + 1)^3] = 3 \cdot \frac{(\sqrt{x} + 1)^2}{2\sqrt{x}}$$

- Integrating both sides with respect to x :

$$3 \int \frac{(\sqrt{x} + 1)^2}{2\sqrt{x}} dx = (\sqrt{x} + 1)^3 + C_1$$

- Dividing by 3:

$$\int \frac{(\sqrt{x} + 1)^2}{2\sqrt{x}} dx = \frac{1}{3}(\sqrt{x} + 1)^3 + C$$

Final Answer: a. $\frac{3(\sqrt{x}+1)^2}{2\sqrt{x}}$ | b. $\frac{1}{3}(\sqrt{x} + 1)^3 + C$

Marks	Marking Criteria Breakdown
2 Marks (a)	1 mark for chain rule application; 1 mark for correct simplified derivative.
2 Marks (b)	1 mark for relating integral to derivative of Part a; 1 mark for primitive with \$+ C\$.

QUESTION 5 | Sequences and Series — Arithmetic Progression (AP)

[3 Marks]

Question: On the first day of the harvest, an orchard produces 1630 kg of fruit. On the next day, the orchard produces 1605 kg, and the amount produced continues to decrease by the same amount each day. On what day does the daily production first fall below 600 kg?

NESA Reference Sheet / Key Rule:
 n -th term of an Arithmetic Progression: $T_n = a + (n - 1)d$

Detailed Working (NESA Standard Solution):

1. Identify AP parameters:
- First term: $a = 1630$
 - Common difference: $d = 1605 - 1630 = -25$
2. State the general n -th term formula:

$$T_n = 1630 + (n - 1)(-25) = 1630 - 25n + 25 = 1655 - 25n$$

3. Form and solve the inequality $T_n < 600$:

$$1655 - 25n < 600 \Rightarrow -25n < 600 - 1655 \Rightarrow -25n < -1055$$

Reversing the inequality when dividing by negative:

$$n > \frac{-1055}{-25} \Rightarrow n > 42.2$$

4. Conclusion:
- Since n represents discrete days, the daily production first falls below 600 kg on day $n = 43$.
(Verification: $T_{42} = 1630 + 41(-25) = 605$ kg; $T_{43} = 1630 + 42(-25) = 580$ kg < 600 kg).

Final Answer: Day 43 (or the 43rd day)

Marks	Marking Criteria Breakdown
1 Mark	Identifies $a = 1630$, $d = -25$ and sets up expression for T_n .
1 Mark	Sets up inequality $T_n < 600$ and solves for $n > 42.2$.
1 Mark	Correctly concludes Day 43.

QUESTION 6 | Differential Calculus — Curve Sketching & Nature of Stationary Points**[7 Marks]****Question:** Consider the curve $y = x^3 - 6x^2 + 9x + 3$.

- (a) Find the location and nature of any stationary points.
 (b) Find the point of inflection.
 (c) Sketch the curve $y = x^3 - 6x^2 + 9x + 3$ labelling the stationary points, the point of inflection and y -intercepts. The x -intercepts are not required.

NESA Reference Sheet / Key Rule:Stationary points: $y' = 0$. Nature: $y'' < 0$ (Local Max), $y'' > 0$ (Local Min). Inflection: $y'' = 0$ with concavity change.**Detailed Working (NESA Standard Solution):****(a) Stationary Points and Their Nature:**

- First derivative:

$$y' = \frac{d}{dx}y = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x-1)(x-3)$$

- Stationary points occur where $y' = 0$:

$$3(x-1)(x-3) = 0 \Rightarrow x = 1 \quad \text{or} \quad x = 3$$

- Corresponding y -coordinates:

- At $x = 1$: $y = (1)^3 - 6(1)^2 + 9(1) + 3 = 1 - 6 + 9 + 3 = 7 \Rightarrow (1, 7)$
- At $x = 3$: $y = (3)^3 - 6(3)^2 + 9(3) + 3 = 27 - 54 + 27 + 3 = 3 \Rightarrow (3, 3)$

- Nature via the second derivative:

$$y'' = \frac{d^2}{dx^2}y = 6x - 12$$

- At $x = 1$: $y''(1) = 6(1) - 12 = -6 < 0 \Rightarrow$ **Local Maximum at (1, 7)**
- At $x = 3$: $y''(3) = 6(3) - 12 = +6 > 0 \Rightarrow$ **Local Minimum at (3, 3)**

(b) Point of Inflection:

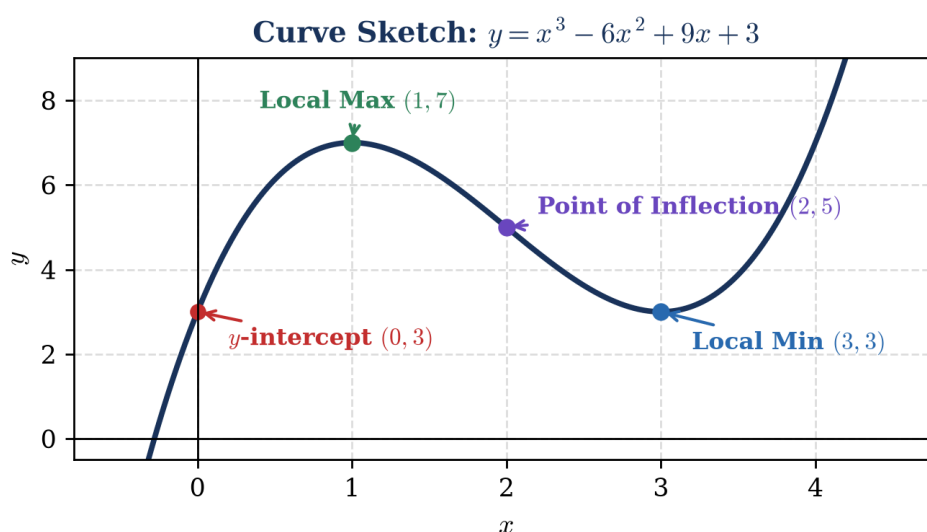
- Potential inflection points occur where $y'' = 0$:

$$6x - 12 = 0 \Rightarrow 6x = 12 \Rightarrow x = 2$$

- At $x = 2$: $y = (2)^3 - 6(2)^2 + 9(2) + 3 = 8 - 24 + 18 + 3 = 5 \Rightarrow (2, 5)$
- Concavity test:
 - For $x < 2$ (e.g. $x = 1$): $y''(1) = -6 < 0$ (concave downwards)
 - For $x > 2$ (e.g. $x = 3$): $y''(3) = +6 > 0$ (concave upwards)
- Since concavity changes sign across $x = 2$, **(2, 5) is a true point of inflection.**

(c) Curve Sketch:

- y -intercept: Let $x = 0 \Rightarrow y = 3 \Rightarrow (0, 3)$.

**Final Answer:** (a) Local Maximum at (1, 7); Local Minimum at (3, 3)

(b) Point of Inflection at (2, 5)

(c) See plotted cubic curve below.

QUESTION 7 | Calculus — Tangents & Normals to Logarithmic Curves

[3 Marks]

Question: A curve has the equation $y = \log_e(x^3 - 5)$. Find the equation of the normal to the curve at the point $x = 2$.

NESA Reference Sheet / Key Rule:
 $\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}$, $m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}$, $y - y_1 = m(x - x_1)$

Detailed Working (NESA Standard Solution):

1. **Find coordinates of the point:** At $x = 2$: $y = \ln(2^3 - 5) = \ln(8 - 5) = \ln 3$. The point is $(2, \ln 3)$.
2. **Differentiate to find tangent gradient:**

$$\frac{dy}{dx} = \frac{\frac{d}{dx}(x^3 - 5)}{x^3 - 5} = \frac{3x^2}{x^3 - 5}$$

At $x = 2$:

$$m_{\text{tangent}} = \frac{3(2)^2}{2^3 - 5} = \frac{12}{3} = 4$$

3. **Determine normal gradient:**

$$m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}} = -\frac{1}{4}$$

4. **Establish equation of normal line:**

$$\begin{aligned} y - \ln 3 &= -\frac{1}{4}(x - 2) \Rightarrow 4(y - \ln 3) = -(x - 2) \\ 4y - 4 \ln 3 &= -x + 2 \Rightarrow x + 4y - (2 + 4 \ln 3) = 0 \quad \text{or} \quad y = -\frac{1}{4}x + \frac{1}{2} + \ln 3 \end{aligned}$$

Final Answer: $x + 4y - (2 + 4 \ln 3) = 0$ ($y = -\frac{1}{4}x + \frac{1}{2} + \ln 3$)

Marks	Marking Criteria Breakdown
1 Mark	Finds point $(2, \ln 3)$ and calculates $\frac{dy}{dx} = \frac{(3x^2)}{(x^3 - 5)}$.
1 Mark	Calculates $m_{\text{tangent}} = 4$ and $m_{\text{normal}} = -1/4$.
1 Mark	Forms correct exact equation of the normal.

QUESTION 8 | Trigonometric Functions — Quadratic Equations**[3 Marks]****Question:** Solve $2\cos^2 x + 3\cos x + 1 = 0$ for $0 \leq x \leq 2\pi$.**NESA Reference Sheet / Key Rule:**Factorise quadratic in $\cos x$; solve in radians across quadrants in $[0, 2\pi]$.**Detailed Working (NESA Standard Solution):****1. Factorise the trigonometric quadratic:**

$$(2\cos x + 1)(\cos x + 1) = 0$$

2. Solve each factor:

$$\cos x = -\frac{1}{2} \quad \text{or} \quad \cos x = -1$$

3. Determine radian solutions in $[0, 2\pi]$:

- **Case 1:** $\cos x = -1 \Rightarrow x = \pi$
- **Case 2:** $\cos x = -\frac{1}{2}$
 - Reference angle: $\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$
 - Cosine is negative in Quadrants II and III:

$$\text{Quadrant II: } x = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\text{Quadrant III: } x = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

Final Answer: $x = \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$

Marks	Marking Criteria Breakdown
1 Mark	Factorises quadratic correctly to $(2\cos x + 1)(\cos x + 1) = 0$.
1 Mark	Obtains boundary solution $x = \pi$ from $\cos x = -1$.
1 Mark	Obtains both exact quadrant solutions $x = (2\pi)/3$ and $x = (4\pi)/3$.

QUESTION 9 | Functions — Absolute Value Equations**[2 Marks]****Question:** Solve the equation $|2x + 3| = 7$.**NESA Reference Sheet / Key Rule:** $|X| = a \Rightarrow X = a \quad \text{or} \quad X = -a \quad (a \geq 0)$ **Detailed Working (NESA Standard Solution):****1. By definition of absolute value, split into two linear cases:**

$$2x + 3 = 7 \quad \text{or} \quad 2x + 3 = -7$$

2. Solve each linear equation:

- **Case 1:** $2x = 7 - 3 = 4 \Rightarrow x = 2$
- **Case 2:** $2x = -7 - 3 = -10 \Rightarrow x = -5$

Final Answer: $x = 2 \quad \text{or} \quad x = -5$

Marks	Marking Criteria Breakdown
1 Mark	Splits absolute value into two cases: $2x + 3 = \text{plus/minus } 7$.
1 Mark	Calculates both correct solutions $x = 2$ and $x = -5$.

QUESTION 10 | Calculus — Product and Chain Rules

[3 Marks]

Question: Find the derivative of $y = \sin 5x \tan(e^x)$.

NESA Reference Sheet / Key Rule:
 $(uv)' = u'v + uv'$, $\frac{d}{dx}[\sin kx] = k \cos kx$, $\frac{d}{dx}[\tan f(x)] = f'(x) \sec^2 f(x)$

Detailed Working (NESA Standard Solution):

1. **Identify factors for Product Rule:**
- Let $u = \sin 5x \Rightarrow u' = 5 \cos 5x$
 - Let $v = \tan(e^x) \Rightarrow v' = \sec^2(e^x) \cdot \frac{d}{dx}(e^x) = e^x \sec^2(e^x)$
2. **Apply product rule formula:**

$$\frac{dy}{dx} = u'v + uv' = 5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$$

Final Answer: $\frac{dy}{dx} = 5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$

Marks	Marking Criteria Breakdown
1 Mark	Correct application of product rule.
1 Mark	Correctly differentiates either $\sin$ or $\tan$ using the chain rule.
1 Mark	Obtains complete, correct derivative expression.

QUESTION 11 | Functions — Composite Functions (Domain and Range)

[3 Marks]

Question: Given $f(x) = \sqrt{x+1}$ and $g(x) = 2x-3$. Find the domain and range of $f(g(x))$.

NESA Reference Sheet / Key Rule:
Radicand of square root must be non-negative (≥ 0); principal square root produces non-negative outputs (≥ 0).

Detailed Working (NESA Standard Solution):

1. **Form composite function:**
- $$f(g(x)) = f(2x-3) = \sqrt{(2x-3)+1} = \sqrt{2x-2}$$
2. **Determine Domain:** For a real square root, the expression under the radical must be non-negative:
- $$2x-2 \geq 0 \Rightarrow 2x \geq 2 \Rightarrow x \geq 1 \quad \text{or} \quad [1, \infty)$$
3. **Determine Range:**
- At $x = 1$: $f(g(1)) = \sqrt{2(1)-2} = \sqrt{0} = 0$.
 - As $x \rightarrow \infty$, $2x-2 \rightarrow \infty \Rightarrow \sqrt{2x-2} \rightarrow \infty$.
 - Since $\sqrt{\cdot}$ denotes the principal (non-negative) square root:

$$y \geq 0 \quad \text{or} \quad f(g(x)) \geq 0 \quad \text{or} \quad [0, \infty)$$

Final Answer: Domain: $x \geq 1$ ($[1, \infty)$) | Range: $y \geq 0$ ($[0, \infty)$)

Marks	Marking Criteria Breakdown
1 Mark	Forms composite expression $f(g(x)) = \sqrt{2x-2}$.
1 Mark	Finds correct domain $x \geq 1$.
1 Mark	Finds correct range $y \geq 0$.

QUESTION 12 | Statistical Analysis — Bivariate Data, Linear Regression & Extrapolation

[6 Marks]

Question: Tom surveyed age (x) and weekly wage (W):

Age (years) (x)	18	45	28	15	32	68
Wage (\$/week) (W)	715	2350	1530	438	1690	1320

- i. Using your calculator, find correlation coefficient (r), and explain strength.
- ii. Find least-squares regression line in form $W = Bx + A$, where A and B are integers.
- iii. Could your equation be used to make valid estimate for ages > 68 or < 15 years? Justify.

NESA Reference Sheet / Key Rule:
 r = Pearson's correlation coefficient, $W = Bx + A$ least-squares line

Detailed Working (NESA Standard Solution):

Part i (Correlation coefficient):

- From 2-variable calculator linear statistics:

$r \approx 0.52632 \approx 0.53$

- Interpretation:** There is a **moderate positive linear correlation** between age and weekly wage.

Part ii (Least-squares regression line):

- Slope: $B = \frac{n\sum xW - \sum x \sum W}{n\sum x^2 - (\sum x)^2} = 18.4795 \Rightarrow B = 18$ (nearest integer).
- Intercept: $A = \bar{W} - B\bar{x} = 706.038 \Rightarrow A = 706$ (nearest integer).
- Regression equation:

$W = 18x + 706$

Part iii (Validity / Extrapolation):

- Answer:** No, the equation cannot be used to make valid estimates.
- Justification:**
 - 1. **Statistical Extrapolation:** Predicting outside the observed sample range $[15, 68]$ is extrapolation. Extrapolation is inherently unreliable because a linear trend cannot be assumed to continue indefinitely.
 - 2. **Contextual Reasons:**
 - For ages > 68 , individuals commonly retire and their wage income decreases significantly; the linear equation falsely predicts continuously increasing earnings.
 - For ages < 15 , individuals are school children not legally working full-time; yet the model unrealistically predicts a wage of \$706 per week at age 0 ($x = 0$).

Final Answer: i. $r \approx 0.53$ (moderate positive linear correlation)
ii. $W = 18x + 706$
iii. No; extrapolation outside $[15, 68]$ is unreliable and unrealistic in reality.

Marks	Marking Criteria Breakdown
2 Marks (i)	1 mark for r approx 0.53; 1 mark for describing as moderate positive linear correlation.
2 Marks (ii)	1 mark for finding slope & intercept; 1 mark for integer equation $W = 18x + 706$.
2 Marks (iii)	1 mark for identifying extrapolation outside $[15, 68]$; 1 mark for valid practical justification.

QUESTION 13 | Calculus / Differential Equations — Exponential Growth**[4 Marks]**

Question: The average life cycle of an insect is one month. The population P grows so that:

$$\frac{dP}{dt} = 1200e^{0.3t}$$

A nest had a population of 5000 after one month ($t = 1$). Determine how long it will take the nest to reach the viable stage of at least 100000. Answer correct to the nearest month.

NESA Reference Sheet / Key Rule:

$$\int e^{kt} dt = \frac{1}{k}e^{kt} + c$$

Detailed Working (NESA Standard Solution):

1. **Integrate the differential rate equation:**

$$P(t) = \int 1200e^{0.3t} dt = \frac{1200}{0.3}e^{0.3t} + C = 4000e^{0.3t} + C$$

2. **Use initial condition $t = 1, P(1) = 5000$ to solve for C :**

$$5000 = 4000e^{0.3(1)} + C \Rightarrow C = 5000 - 4000e^{0.3}$$

$$C \approx 5000 - 4000(1.349859) = 5000 - 5399.435 = -399.435$$

3. **State the population function:**

$$P(t) = 4000e^{0.3t} - 399.435$$

4. **Solve for t when $P(t) = 100000$:**

$$4000e^{0.3t} - 399.435 = 100000 \Rightarrow 4000e^{0.3t} = 100399.435$$

$$e^{0.3t} = \frac{100399.435}{4000} = 25.09986$$

Taking the natural logarithm:

$$0.3t = \ln(25.09986) \approx 3.22286 \Rightarrow t = \frac{3.22286}{0.3} \approx 10.74 \text{ months}$$

5. **Round to nearest month: $t \approx 11$ months.**

Final Answer: 11 months

Marks	Marking Criteria Breakdown
1 Mark	Integrates correctly to obtain $P(t) = 4000e^{(0.3t)} + C$.
1 Mark	Substitutes $t = 1, P = 5000$ to find $C = 5000 - 4000e^{(0.3)}$ approx -399.44\$.
1 Mark	Sets $P(t) = 100\,000$ and solves for t approx 10.74\$.
1 Mark	Rounds correctly to the nearest month (11 months).

QUESTION 14 | Continuous Probability — Normal Distribution**[3 Marks]**

Question: Weights of bags of red gravel are modelled by a normal distribution with mean 25.8 kg and standard deviation 0.5 kg. Using the provided standard normal cumulative table, find the probability that a bag weighs between 25.5 kg and 26.5 kg.

NESA Reference Sheet / Key Rule:

$$Z = \frac{X - \mu}{\sigma}, \quad P(a \leq X \leq b) = \Phi(z_2) - \Phi(z_1), \quad \Phi(-z) = 1 - \Phi(z)$$

Detailed Working (NESA Standard Solution):**1. Calculate z -scores for the boundary values ($\mu = 25.8, \sigma = 0.5$):**

- For $x_1 = 25.5$ kg: $z_1 = \frac{25.5 - 25.8}{0.5} = \frac{-0.3}{0.5} = -0.6$
- For $x_2 = 26.5$ kg: $z_2 = \frac{26.5 - 25.8}{0.5} = \frac{0.7}{0.5} = 1.4$

2. Express probability via the standard normal cumulative function $\Phi(z)$:

$$P(25.5 \leq X \leq 26.5) = P(-0.6 \leq Z \leq 1.4) = \Phi(1.4) - \Phi(-0.6)$$

3. Extract probabilities from provided table:

- For $z = 1.4$: Row 1., Col .4 $\Rightarrow \Phi(1.4) = 0.9192$.
- For $z = -0.6$: By symmetry of standard normal distribution:

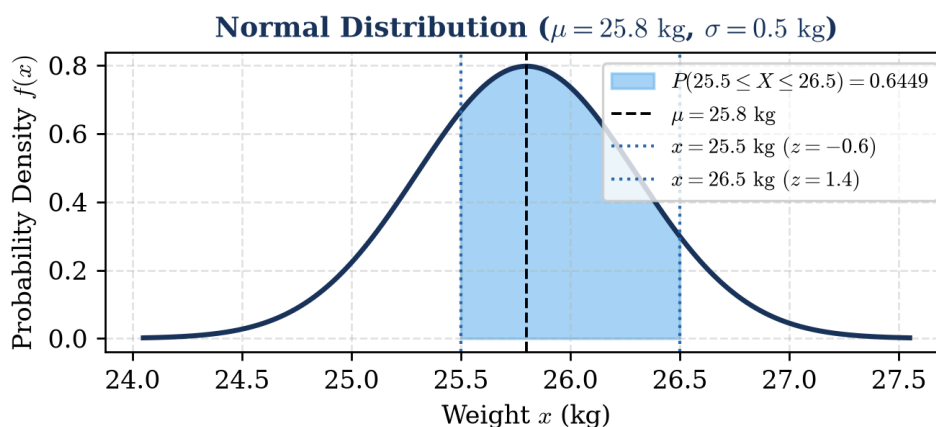
$$\Phi(-0.6) = 1 - \Phi(0.6)$$

From table (Row 0., Col .6): $\Phi(0.6) = 0.7257$.

$$\Phi(-0.6) = 1 - 0.7257 = 0.2743$$

4. Compute final probability:

$$P(-0.6 \leq Z \leq 1.4) = 0.9192 - 0.2743 = 0.6449 \quad (64.49\%)$$



Final Answer: 0.6449 (64.49%)

Marks	Marking Criteria Breakdown
1 Mark	Standardises both values to $z_1 = -0.6$ and $z_2 = 1.4$.
1 Mark	Extracts $\Phi(1.4) = 0.9192$ and computes $\Phi(-0.6) = 1 - 0.7257 = 0.2743$ from table.
1 Mark	Evaluates final difference 0.6449 .

QUESTION 15 | Calculus / Optimization — Mensuration & Cost Minimization**[6 Marks]**

Question: Harvey is designing an open-ended tent as shown below. The frame is made using rods of two lengths:

- x metres for top and bottom edges
- y metres for each sloping edge.

Rods cost \$9.50 per metre. The frame is covered by one rectangular sheet of fabric of area 24m^2 .

- Show that the total length, l m, of rods is given by $l = 3x + \frac{48}{x}$.
- Find the value of x for which l is a minimum.
- What is the minimum cost of the frame?

NESA Reference Sheet / Key Rule:

Fabric area = $2xy = 24$. Stationary point: $\frac{dl}{dx} = 0$. Minimum test: $\frac{d^2l}{dx^2} > 0$.

Detailed Working (NESA Standard Solution):

(a) Show that $l = 3x + \frac{48}{x}$:

- Rods of length x : 1 top horizontal ridge + 2 bottom horizontal ground edges = 3 rods of length x .
- Rods of length y : 2 sloping rods at front triangular opening + 2 at back opening = 4 rods of length y .
- Total length of rods:

$$l = 3x + 4y$$

- The fabric is one continuous rectangular sheet draped over the ridge covering both sloping roofs:
 - Sheet length = x , sheet width = $y + y = 2y$.
 - Fabric Area = $x(2y) = 2xy = 24 \Rightarrow y = \frac{24}{2x} = \frac{12}{x}$.
- Substituting y into l :

$$l = 3x + 4\left(\frac{12}{x}\right) = 3x + \frac{48}{x} \quad (\text{Shown})$$

(b) Find x for which l is a minimum:

- Differentiate l with respect to x :

$$\frac{dl}{dx} = \frac{d}{dx}(3x + 48x^{-1}) = 3 - 48x^{-2} = 3 - \frac{48}{x^2}$$

- Stationary points occur where $\frac{dl}{dx} = 0$:

$$3 - \frac{48}{x^2} = 0 \Rightarrow 3x^2 = 48 \Rightarrow x^2 = 16 \Rightarrow x = 4 \text{ m} \quad (\text{since } x > 0)$$

- Verify minimum using the second derivative test:

$$\frac{d^2l}{dx^2} = \frac{d}{dx}(3 - 48x^{-2}) = 96x^{-3} = \frac{96}{x^3}$$

At $x = 4$:

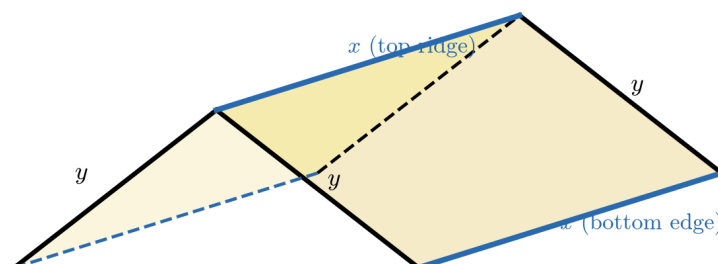
$$\frac{d^2l}{dx^2} = \frac{96}{4^3} = \frac{96}{64} = 1.5 > 0$$

Since $\frac{d^2l}{dx^2} > 0$, the total length l is a **minimum at $x = 4$ m**.

(c) **Minimum cost of the frame:**

- Total minimum rod length: $l(4) = 3(4) + \frac{48}{4} = 12 + 12 = 24$ m.
- Total cost:

$$\text{Cost} = 24 \text{ m} \times \$\frac{9.50}{\text{m}} = \$228.00$$

Open-Ended Triangular Prism Tent Frame

QUESTION 16 | Financial Mathematics — Future Value of Annuities

[3 Marks]

Question: A table of future value interest factors for an annuity of \$1 is shown:

Period	1%	4%	8%	12%	16%	20%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0400	2.0800	2.1200	2.1600	2.2000
3	3.0301	3.1216	3.2464	3.3744	3.5056	3.6400
4	4.0604	4.2465	4.5061	4.7793	5.0665	5.3680
5	5.1010	5.4163	5.8666	6.3528	6.8771	7.4416
6	6.1520	6.6330	7.3359	8.1152	8.9775	9.9299

Annie wants to save \$10000 over 2 years by investing equal payments every 6 months at 8% p.a. compounded half-yearly. What should be the value of each payment?

NESA Reference Sheet / Key Rule:
Future Value = Payment (M) $\times$ Table Factor, n = periods, r = rate per period

Detailed Working (NESA Standard Solution):

1. Determine number of periods (n) and interest rate per period (r):
- Total time = 2 years.
 - Payment frequency = every 6 months (half-yearly).
 - Compounding periods: $n = 2 \times 2 = 4$ periods.
 - Interest rate per half-year: $r = \frac{8\% \text{ p.a.}}{2} = 4\%$ per half-year.
2. Extract the factor from the provided table:
- Row: Period = 4
 - Column: 4%
 - Table Factor = 4.2465
3. Calculate payment amount (M):

Future Value = $M \times 4.2465$

$\$10000 = M \times 4.2465 \Rightarrow M = \frac{10000}{4.2465} = 2354.880489...$

4. Round to nearest cent: $M = \$2354.88$.

Final Answer: \$2354.88

Marks	Marking Criteria Breakdown
1 Mark	Identifies $n = 4$ periods and $r = 4\%$ rate per period.
1 Mark	Extracts correct factor 4.2465 from the annuity table.
1 Mark	Calculates payment $\$M = \2354.88 rounded to the nearest cent.

Summary of All Final Answers (Questions 1 to 16)

Quick Reference Table Aligned to NESA Marking Standards

Q#	Topic / Syllabus Area	Final Answer (NESA Standard)
1	Calculus — Linear Substitution	$\frac{1}{6}(4x + 1)^{\frac{3}{2}} + C$
2	Calculus — Increasing Functions	$x \leq -\sqrt{3}$ or $x \geq \sqrt{3}$ $((-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty))$
3	Calculus — Logarithmic Integration	$\frac{1}{3} \ln(e^{3x} + 1) + C$
4a	Calculus — Chain Rule	$\frac{3(\sqrt{x}+1)^2}{2\sqrt{x}}$
4b	Calculus — Reverse Chain Rule	$\frac{1}{3}(\sqrt{x} + 1)^3 + C$
5	Sequences & Series — AP	Day 43
6a	Calculus — Stationary Points	Local Max: (1, 7) Local Min: (3, 3)
6b	Calculus — Point of Inflection	Point of Inflection: (2, 5)
6c	Calculus — Curve Sketching	Smooth cubic through (0, 3), max at (1, 7), inflection at (2, 5), min at (3, 3)
7	Calculus — Tangents & Normals	$x + 4y - (2 + 4 \ln 3) = 0$ ($y = -\frac{1}{4}x + \frac{1}{2} + \ln 3$)

8	Trigonometry — Quadratic Eqn	$x = \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$
9	Functions — Absolute Value	$x = 2$ or $x = -5$
10	Calculus — Product & Chain Rule	$5 \cos 5x \tan(e^x) + e^x \sin 5x \sec^2(e^x)$
11	Functions — Domain & Range	Domain: $x \geq 1$ ($[1, \infty)$) Range: $y \geq 0$ ($[0, \infty)$)
12i	Statistics — Correlation	$r \approx 0.53$ (moderate positive linear correlation)
12ii	Statistics — Regression Line	$W = 18x + 706$
12iii	Statistics — Validity	No; extrapolation outside $[15, 68]$ is unreliable & unrealistic in reality
13	Differential Eqn — Exponential	11 months
14	Normal Distribution	0.6449 (or 64.49%) satisfying $P(-0.6 \leq Z \leq 1.4)$
15a	Optimization — Total Length	Shown: $l = 3x + \frac{48}{x}$
15b	Optimization — Minimum Length	$x = 4$ m
15c	Optimization — Minimum Cost	\$228.00
16	Financial Maths — Annuities	\$2354.88